

All of the orbitals in a given shell have the same value for their _____ quantum number.

Select one:

- ☐ Secondary (Angular) (L)
- ☐ Spin (ms)
- ☒ Principal (n)
- ☐ Magnetic (mL)

How many unpaired electrons are found in V (atomic number = 23)

Select one:

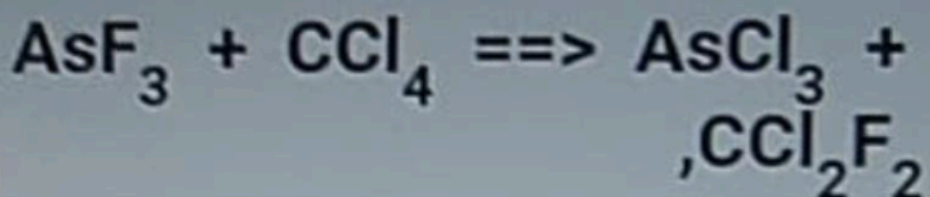
☐ 0

☐ 1

☒ 3

☐ 4

in the reaction



If the theoretical yield of CCl_2F_2 was 1.68 moles, and the percent yield in the reaction was 74.3%, how many grams of CCl_2F_2 (121 g/mol) were actually obtained

اختر أحد الخيارات

151 ☒

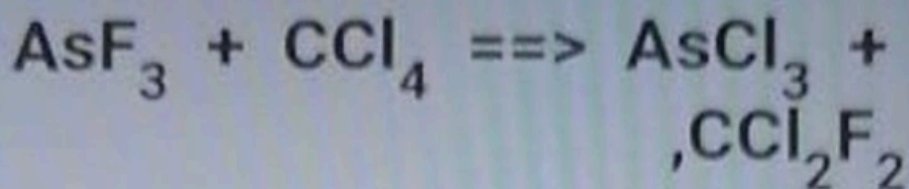
203 ☐

303 ☐

167 ☐

273 ☐

In the reaction



If the theoretical yield of CCl_2F_2 was 1.68 moles, and the percent yield in the reaction was 74.3%, how many grams of CCl_2F_2 were actually obtained

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303

203

167

273

151

How many different orientations in space do s-orbitals have?

Select one:

☐ 5

☐ 2

☐ 7

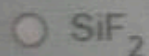
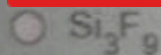
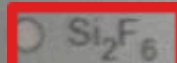
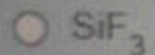
☒ 1

☐ 3

A compound was found to contain 33.0% Si and 67.0% F, by mass. If the molar mass of the compound is 170.0 g/mol, find its molecular formula.

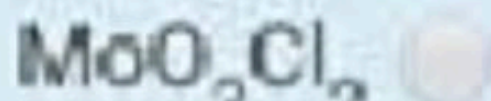
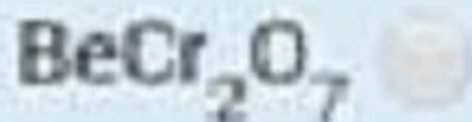
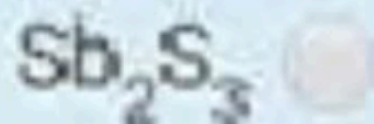
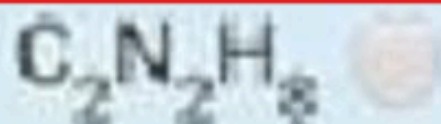
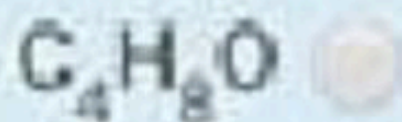
(Si = 28 , F = 19 g/mol)

Select one:



Which of the following is
?not an empirical formula

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percentage of fluorine (F) in a compound with the empirical formula SF_6 ?

S = 32, F = 19 g/mol

Select one:

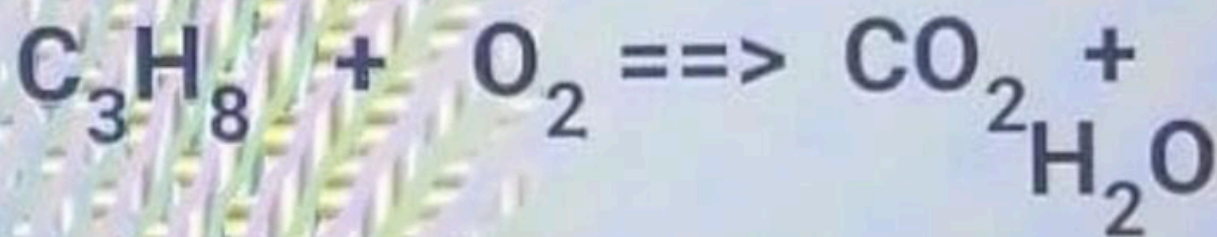
☐ 29.67%

☐ 36.00%

☐ 21.95%

☒ 78.08%

The correct coefficient that balanced The O_2 in the chemical equation for the reaction below



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12

19

14

7

5

:chemical equation



If 0.36 moles of C_4H_4 are allowed to react with 2.82 moles of O_2 , what is the mass of water produced (in grams)

molar mass of water = 18 g/mol

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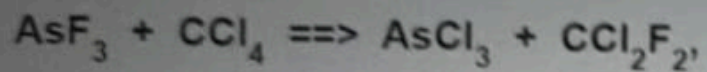
13.20 ☐

19.64 ☐

12.96 ☒

65.50 ☐

11.02 ☐

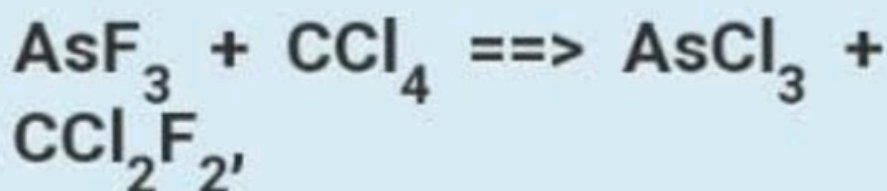


If the theoretical yield of CCl_2F_2 was 1.68 moles, and the percent yield in the reaction was 74.3% grams of CCl_2F_2 (121 g/mol) were actually obtained?

Select one:

- ☐ 273
- ☒ 167
- ☐ 303
- ☐ 151
- ☐ 203

In the reaction

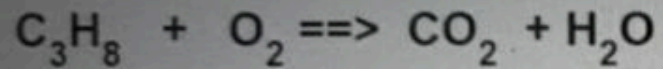


If the theoretical yield of CCl_2F_2 was 1.68 moles, and the percent yield in the reaction was 82.2 %, how many grams of CCl_2F_2 (121 g/mol) were actually obtained?

Select one:

- ☒ 167
- ☐ 203
- ☐ 273
- ☐ 151
- ☐ 303

below.



Select one:

☐ 19

☐ 12

☐ 14

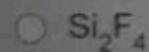
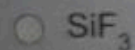
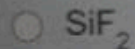
☒ 5

☐ 7

A compound was found to contain 33.0% Si and 67.0% F, by mass. If the molar mass of the compound is 170.0 g/mol, find its molecular formula.

(Si = 28 , F = 19 g/mol)

Select one:



The lowest energy "f" subshell is : ____^f

Select one:

☒ 4

☐ 2

☐ 5

☐ 3

☐ 1

Arrange the following atoms according to decrease in their ionization energies

Al (At.no. = 13)

Ga (At.no. = 31)

Mg (At.no. = 12)

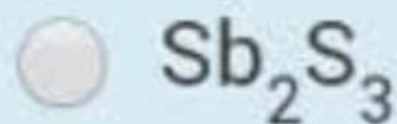
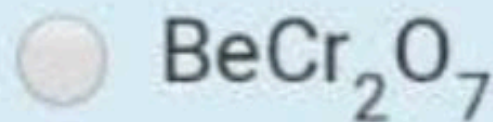
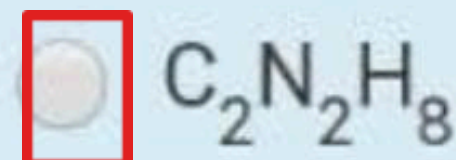
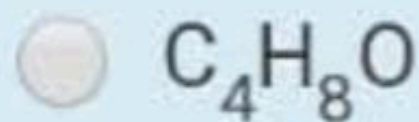
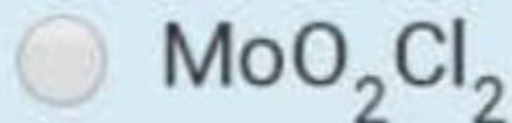
Ar (At.no. = 18)

Select one:

- ☒ Ar > Mg > Al > Ga
- ☐ Mg > Al > Ga > Ar
- ☐ Ga > Mg > Al > Ar
- ☐ Ar > Al > Mg > Ga
- ☐ Al > Mg > Ar > Ga

Which of the following is not an empirical formula?

Select one:



How many molecules of (C_4H_8O) are present in

2.00×10^{-8} mol of (C_4H_8O) ?

Avogadro's number = 6.022×10^{23}

C = 12, H=1, O=16 g/mol

Select one:

☐ 1.67×10^{13}

☒ 1.20×10^{16}

☐ 1.20×10^{15}

☐ 1.67×10^{18}

☐ 1.67×10^{14}

Down the group (from top to bottom) the electron affinity _____

Select one:

- ☐ remains constant because effective nuclear charge is constant
- ☐ increases due to increase of number of shells
- ☐ decreases then increases
- ☒ decreases because number of shells increase
- ☐ increase because effective nuclear charge increases

Avogadro's number = 6.022×10^{23}

C = 12, H=1, O=16 g/mol

Select one:

- ☐ 0.64 mol
- ☐ 0.58 mol
- ☐ 0.78 mol
- ☒ 0.064 mol
- ☐ 0.081 mol

How many different orientations in space do s-orbitals have?

Select one:

☐ 5

☐ 2

☐ 7

☒ 1

☐ 3